

Kshitiz International College

SEND UP EXAMINATIONS

Level: Master Trimester: Spring Year : 2026
Programme: MBA/MBA(Finance)/MBA(GB) Full Marks: 100
Course: Management Information System Pass Marks: 60
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Section 'A'

Conceptual Questions

*Attempt **any five** questions from this section* [5×8=40]

- 1) Discuss the role of MIS in planning, controlling and decision making activities of modern organizations with data interpretations.
- 2) Differentiate between Transaction Processing System, Management Information System, Decision Support System and Executive Support System with suitable business examples.
- 3) Explain how database management systems support business operations and managerial decision making.
- 4) Discuss the role of cloud computing, Internet of Things and blockchain in modern business information systems.
- 5) Discuss the importance of privacy, data protection and legal compliance in Management Information Systems.
- 6) Explain the concepts of data warehouse, data mining and business intelligence and discuss their role in decision making.

Section 'B'

Analytical Questions

*Attempt **any two** questions from this section* [2×15=30]

- 7) Evaluate the strategic impact of emerging technologies such as Artificial Intelligence, Cloud Computing, Internet of Things and Blockchain in modern business organizations. Support your answer with suitable case examples with data.
- 8) Suppose a company is collecting large amounts of customer data from sales transactions, website visits, mobile apps and social

media. Explain how data warehousing, data mining, business intelligence and big data analytics can help managers make better business decisions.

- 9) A commercial bank is facing phishing attacks, unauthorized employee access and customer data privacy complaints. Analyze the cyber security, ethical and legal issues involved and suggest suitable security controls, compliance measures and audit procedures to reduce risk.

Section ‘C’

Comprehensive Questions

Read the context given below and answer the questions that follow: [30]

- 10) NepalConnect Telecom is one of the growing telecommunication companies in Nepal. It provides mobile voice service, fiber internet, digital TV and corporate connectivity solutions to individual and business customers. For many years, the company had a stable customer base because there were limited service providers in the market. Recently, competition has increased rapidly. New internet service providers, mobile data companies and digital communication platforms are offering cheaper packages, faster internet and better customer service. As a result, NepalConnect has started losing customers every month.

The management team investigated the major reasons behind customer churn. They found that many customers were leaving because of high monthly charges, frequent network downtime, slow complaint handling, unclear billing and attractive competitor offers. Some customers complained that their service problems were not solved on time. Others said that the company did not provide personalized packages according to their usage pattern. Although NepalConnect had large amounts of customer data, the data was stored in different systems. Billing data was maintained by the finance department, complaint records were stored in the customer support system, network downtime data was handled by the technical department and customer profiles were stored in the CRM system. Because these systems were not integrated, managers could not easily identify customers who were at risk of leaving.

To solve this problem, NepalConnect planned to implement an integrated Management Information System. The system would

combine customer profile, billing history, service usage, complaint records, satisfaction score, contract type, payment behavior, network downtime and competitor offer information. At the operational level, the system would record daily transactions such as bill payment, service activation, complaint registration and package changes. At the managerial level, the system would generate reports on churn rate, unresolved complaints, region wise service problems, customer satisfaction and revenue loss. The company also planned to use Business Intelligence and predictive analytics to classify customers into low, medium and high churn risk groups.

The proposed system could help managers design better retention strategies. For example, high value customers with frequent complaints could receive priority support. Customers with high monthly charges could be offered customized packages. Customers affected by frequent downtime could receive compensation or service upgrades. However, the system also raised important concerns. NepalConnect would collect and analyze sensitive customer information such as phone number, payment history, location based usage, complaint details and service behavior. If such data was accessed by unauthorized employees, leaked to third parties or used for excessive marketing without consent, customer trust could be damaged. Therefore, the company needed strong cybersecurity, privacy policy, role based access control, audit logs, customer consent and ethical data use guidelines before fully implementing the system.

- a)** Explain how an integrated Management Information System can help NepalConnect Telecom reduce customer churn and improve managerial decision making.
- b)** Identify how TPS, MIS, DSS and ESS can be used in the proposed telecom information system.
- c)** Analyze how Business Intelligence, predictive analytics, CRM and cloud computing can support NepalConnect's customer retention strategy.
- d)** Discuss the major cybersecurity, privacy and ethical issues raised by the use of customer data in this case.

- e) Suggest suitable managerial and technical controls that NepalConnect should implement to protect customer data and improve customer trust.